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NASCAR Sponsorship Visibility Analysis

Final Report for NY Racing

Prepared by: Lloyd Todaro Date: August 18, 2026 Version: 1.0

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Section 1: Executive Summary

Executive Summary

Overview

This report presents a comprehensive analysis of NASCAR Cup Series sponsorship visibility for NY Racing's strategic planning. Using data from the 2025 season and 2026 season-to-date, we developed a quantitative visibility scoring model to compare sponsorship opportunities and identify the best value investments.

Scope: 5 primary sponsors analyzed across 53 tracked races (each sponsor's own race count ranges from 11 to 29, reflecting races it actually carried as primary sponsor)

Methodology: Composite visibility score incorporating race performance (40%), a PCA-weighted visibility index across Reddit/YouTube/news (50%), and special events: wins and playoffs (10%)

Key Findings

Finding 1: Efficiency leadership doesn't track brand prestige

Cheddar's Scratch Kitchen (Kyle Busch, Richard Childress Racing) delivers the best visibility-per-dollar of any tracked sponsor (roughly 1.5x the next-best sponsor) while carrying a budget comparable to Castrol's and far below Progressive's. It isn't a "cheap but small" pick either: it's #2 in total visibility, trailing only Progressive, which costs roughly 3.7x as much.

Supporting evidence: Efficiency 39.9 pts/\$M vs. 26.5 pts/\$M for Progressive, the next-best (efficiency rank #1 of 5). Total visibility 195.6 points (rank #2 of 5) on an estimated \$4.9M budget.

Finding 2: A playoff race is worth more than double a regular-season race

The visibility bump from playoff races isn't a marginal effect. A playoff race is worth more than double a regular-season one, a bigger swing than the win bonus (1.79x) and second only to the top-5 bonus (3.12x). Sponsors should treat playoff-window races as prime, high-value activation windows, not just "more of the same, but with higher stakes."

Supporting evidence: Average visibility in playoff-window races (race 27-36) is 129% higher than in regular-season races, across 20 playoff sponsor-entries vs. 77 regular-season entries.

Finding 3: Reddit is the only channel with a statistically reliable signal

Of the three visibility channels tracked, only Reddit's relationship with race performance clears standard statistical significance thresholds with a reasonably sized sample. YouTube and news readings are directionally similar but not statistically distinguishable from noise at this sample size. That's a real constraint on how much weight either should carry in future live monitoring.

Supporting evidence: Reddit mentions vs. finish position: $r = -0.34$, $p = 0.0006$, $N = 97$ (the strongest and most significant relationship in the correlation matrix). YouTube views vs. finish position: $r = -0.20$, $p = 0.055$ (not significant at the 0.05 level). News mentions vs. finish position: $r = -0.13$, $p = 0.20$ (not significant).

Top-Line Results

Metric	Leader	Score
Highest Visibility	Progressive	477.6 points
Best Efficiency	Cheddar's Scratch Kitchen	39.9 pts/\$M
Visibility Range	All sponsors	71 - 478 points
Efficiency Range	All sponsors	13.6 - 39.9 pts/\$M

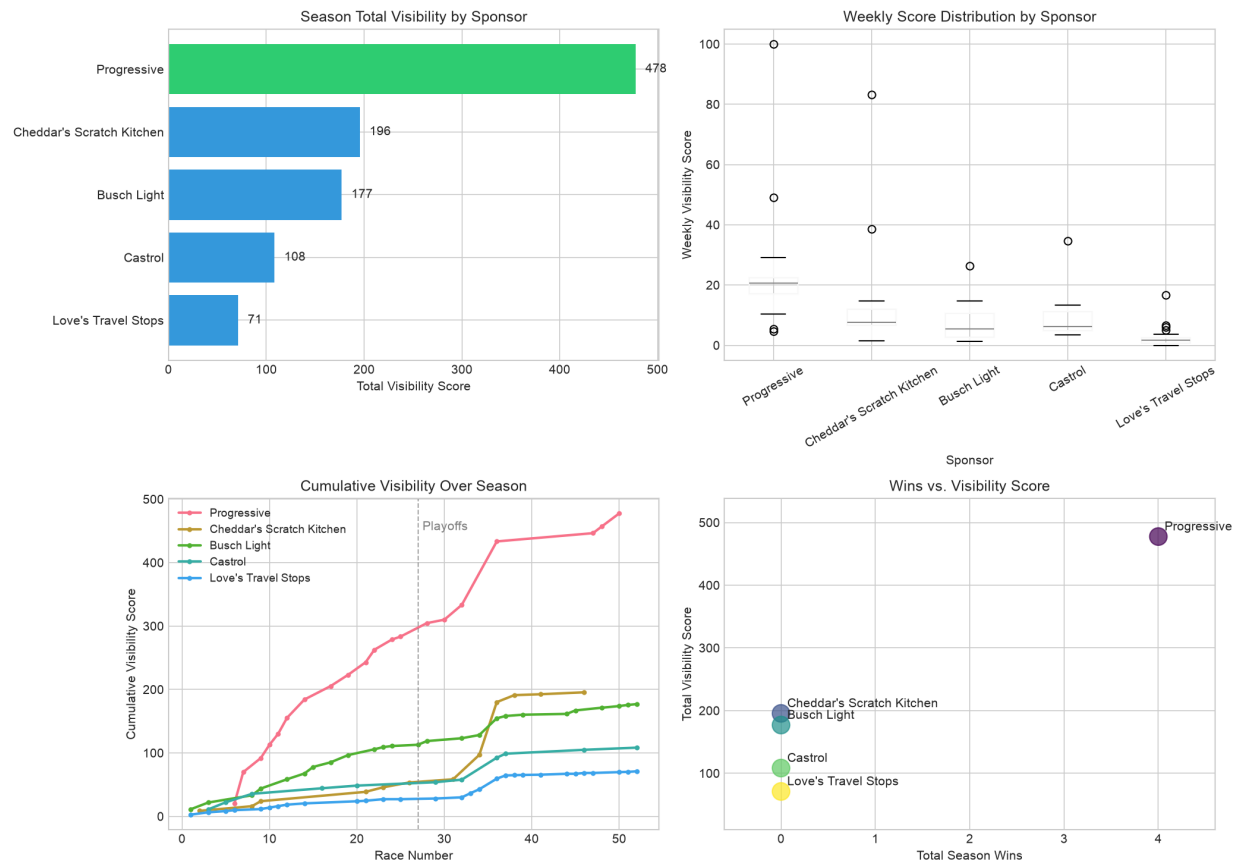


Figure 1: Sponsor visibility dashboard: season totals, weekly score distribution, cumulative trend, and wins vs. visibility

The cumulative-visibility panel (bottom left) shows Progressive separating from the field early and extending the lead again at the playoff mark (Finding 2). The other four sponsors stay bunched together for most of the season, until Cheddar's Scratch Kitchen overtakes Busch Light in the final third, the same efficiency story told numerically in Finding 1. The wins-vs-visibility panel (bottom right) makes that finding visual: Progressive is the only sponsor with any wins, yet Cheddar's Scratch Kitchen still out-visibilityes Castrol and Love's Travel Stops by a wide margin.

Recommendations

Primary Recommendation

Cheddar's Scratch Kitchen is recommended as the primary sponsorship target based on: - Best efficiency of the five sponsors (39.9 pts/\$M, ~1.5x Progressive) - #2 total visibility (195.6 points) despite the smallest

budget in the group - Lowest financial commitment required to secure a top-2 visibility outcome

Estimated investment: \$4.9M (low-medium confidence estimate; verify directly before committing, see caveats) | **Expected visibility:** ~196 points

Alternative Option

Progressive offers the strongest total exposure for teams prioritizing maximum reach over cost efficiency: - #1 total visibility (477.6 points), driven by the strongest on-track performance of any sponsor (4 wins, 11 top-5s in 20 sponsored races) - Best-sourced, medium-confidence cost estimate of the group, but at roughly 3.7x Cheddar's estimated cost

Castrol is a smaller, well-sourced hedge if Cheddar's cost verification comes back unfavorable, though it is no longer a standout on either visibility or efficiency once restricted to its actual 11 sponsored races.

Recommended Next Steps

1. Verify the Cheddar's Scratch Kitchen cost estimate directly (a quote or a better-sourced comparable). It's the single number that could flip this recommendation.
2. Run a sentiment check on the Kyle Busch Reddit/YouTube corpus to confirm the visibility is net-positive before pitching this externally.
3. Revisit the ranking once Cheddar's and Castrol (both under 12 tracked races) have accumulated more races, to confirm the efficiency lead holds up beyond the current small sample.

Important Caveats

This analysis relies on **estimated sponsorship costs** derived from industry publications. Actual costs may vary significantly from estimates. Cheddar's Scratch Kitchen's estimate in particular carries low-medium confidence with no disclosed source for this specific deal. Rankings are sensitive to:

- Cost estimate accuracy (tested with sensitivity analysis): Cheddar's efficiency lead could shrink or disappear if true cost exceeds ~\$7M
- Small sample sizes for Castrol (11 races) and Cheddar's Scratch Kitchen (12 races), which widen the uncertainty on their numbers relative to Progressive (20), Busch Light (25), and Love's Travel Stops (29)
- Weight configuration choices (model is nominally robust to weight changes, though this mostly reflects large gaps between sponsors' scores rather than fine-grained model calibration)
- 2025-2026 season-specific patterns (may not generalize to future seasons)

See Section 6 (Limitations) for complete discussion.

For detailed methodology, see Section 2. For complete findings, see Sections 3-4. For recommendation rationale, see Section 5.

Section 2: Methodology

2.1 Data Sources

This analysis draws on four primary data categories collected for the 2025 NASCAR Cup Series season and 2026 season-to-date.

Race Performance Data

Source	Data Type	Collection Method	Records
Racing Reference	Race results	Web Scrape (BeautifulSoup)	53 races

Fields collected: - Race name, date, track - Driver, team, finish position - Laps led, stage points - DNF/crash flags

Data quality notes: - Web scraper built with BeautifulSoup, pulling race-by-race result pages - No missing race records for tracked sponsors - Primary sponsor identification uses the per-race sponsor field, matched against each tracked driver’s sponsor keywords for that specific race, not a driver’s full-season assignment

Social Media Data

Source	Data Type	Collection Method	API/Tool
Reddit (r/NASCAR)	Post mentions	Arctic Shift	Python script
YouTube	Video engagement	YouTube Data API v3	Python script

Reddit data fields: - Number of posts - Sponsor mention flag

YouTube data fields: - Video title - View count, like count - Upload date - Sponsor relevance flag

Collection period: July 9 - August 16, 2026 (2025 season data collected first; 2026 season-to-date and sponsor-specific supplements collected through mid-August)

Rate limiting: YouTube Data API is quota-limited (~100 units per search call), with short delays paced between per-driver queries. Reddit (Arctic Shift) has no fixed published rate limit; requests use retry-with-backoff to handle intermittent failures.

News Coverage Data

Source	Data Type	Collection Method
GDELT DOC 2.0 API + BigQuery GKG (Global Knowledge Graph)	News article mentions	Automated query, restricted to a curated list of ~19 NASCAR/motorsports and general sports domains (ESPN, NASCAR.com, Fox Sports, Jayski, Frontstretch, Speedway Media, and others), with article text fetched and cached for sponsor keyword matching

Collection window: Monthly windows spanning February-December 2025 and February-December 2026.

Weighting approach (4-tier source weighting): - Tier 1 (ESPN, NASCAR.com, Fox Sports, CBS Sports, NBC Sports): Weight = 1.0 - Tier 2 (Speedway Media, motorsport.com): Weight = 0.8 - Tier 3 (Jayski, Frontstretch, si.com, and any unlisted domain by default): Weight = 0.6 - Tier 4 (reserved for lower-tier blogs/fan sites, none currently mapped): Weight = 0.4

Cost Estimate Data

Source	Data Type	Reliability
Forbes NASCAR coverage	Sponsorship valuations	Medium
Jayski	Deal/partnership reports	Medium
Speedway Media	Partnership announcements	Low-Medium
RTR Sports	Cost benchmarks	Low-Medium
NASCAR.com / NBC Sports / ESPN press coverage	Partnership announcements	Medium

Note: Exact sponsorship costs are confidential. All figures are estimates based on publicly available information.

2.2 Sponsor Selection

Five primary sponsors were selected for analysis based on:

1. **Data availability:** Sufficient race, social, and media data
2. **Team diversity:** Mix of top-tier, mid-tier, and small-team organizations
3. **Sponsor stability:** Consistent primary or near-primary presence across the tracked window
4. **Visibility:** Clear livery presence for attribution

Selected sponsors:

Sponsor	Team	Driver	Team Tier
Progressive	Joe Gibbs Racing	Denny Hamlin	Top
Cheddar's Scratch Kitchen	Richard Childress Racing	Kyle Busch	Mid
Castrol	RFK Racing	Brad Keselowski	Mid-to-top
Love's Travel Stops	Front Row Motorsports	Todd Gilliland	Small/underfunded
Busch Light	Trackhouse Racing	Ross Chastain	Mid

2.3 Visibility Scoring Model

Overview

We developed a composite visibility score that combines three top-level categories into a single metric for sponsor comparison. The model is designed to answer: "How much visibility exposure did this sponsor receive during the season?"

The category weights below (40/50/10) are heavily derived from industry-standard sponsorship valuation practice rather than fit to this dataset.

Score Components and Weights

Category	Weight	Components	Rationale
Race Performance	40%	Laps led (40%)	On-track dominance drives camera time and attention. Laps led is used instead of finish position because the two are highly collinear, and laps led showed the stronger relationship with visibility in EDA. Including both would double-count the same signal.
Visibility Index	50%	Reddit mentions, YouTube views, and weighted news mentions, combined via PCA	Captures total cross-channel exposure using data-driven relative weights between channels, rather than an analyst hand-picking a split between
Special Events	10%	Race wins (6%), Playoff races (4%)	Reddit/YouTube/news Wins and playoff races generate disproportionate visibility spikes that a smooth performance metric alone would miss

Visibility Index Construction

The 50% Visibility Index is itself a composite, built as follows:

1. Each channel (Reddit mention count, YouTube total view count, news weighted mentions) is z-score standardized.
2. Principal Component Analysis (PCA) is fit across the three standardized channels; the first principal component's loadings become the data-driven relative weights between channels.
3. Each channel's loading is additionally scaled by $\sqrt{\text{non-zero sample count} / \text{total sample count}}$, so a channel with much sparser coverage (news: 36 of 223 sponsor-races) doesn't get overweighted relative to a channel with near-complete coverage (Reddit: 222 of 223).
4. The resulting composite is shifted to be non-negative (**Visibility_Score_Shifted**).

Normalization Method

Continuous inputs (laps led and the visibility index) are z-score standardized rather than min-max normalized, so a single outlier race doesn't compress the rest of the scale. Laps led in particular is heavily skewed, with most races recording 0 laps led and a handful recording 200+.

Binary flags (**is_win**, **is_playoff**) are kept as plain 0/1 indicators rather than z-scored, since z-scoring a category flag would rescale it by an arbitrary sample standard deviation rather than add meaningful information.

The weighted composite of all standardized and binary components is calculated first in raw units; the full result is then rescaled to a single 0-100 scale in one final min-max transformation, rather than rescaling each variable individually before combining.

Special Cases

- **Finish position** is tracked in the scoring configuration but carries zero effective weight in the final model, for the collinearity reason described above: laps led carries the race-performance signal instead.
- **Wins** are defined as a 1st-place finish.
- **Playoff races** are races 27-36, matching NASCAR's actual 10-race playoff window.
- All variables are drawn from `scored_dataset.csv`, the output of applying this model to every sponsor-race combination in the tracked window.

2.4 Data Validation and Sensitivity Testing

2.4.1 Model Validation Checks

Three categories of automated checks were run against the scored dataset (`scored_dataset.csv`) before using it for ranking and recommendations.

Performance-visibility correlation (sponsor-season level, N=5 tracked sponsors):

Check	Result	Expected Direction	Status
Average finish position vs. total visibility	$r = -0.985$	Negative	PASS
Total wins vs. total visibility	$r = 0.949$	Positive	PASS
Total laps led vs. total visibility	$r = 0.951$	Positive	PASS

These sponsor-season correlations are a directional sanity check on only 5 aggregated points, and a handful of aggregated points will produce very high correlations almost by construction. The statistically meaningful read is the race-level relationship in `correlation_results.csv` (N = 97): Reddit mentions vs. finish position, $r = -0.34$, $p = 0.0006$. Treat the sponsor-level numbers above as confirmation the model points the right direction, not as model validation in the statistical sense.

Event impact validation (race level):

Check	Result	Status
Wins score meaningfully higher than non-wins	1.79x	PASS
Playoff races score higher than regular-season races	+129% (2.29x)	PASS
Top-5 finishes score higher than finishes outside the top 20	3.12x	PASS

Anomaly detection (6 flagged, all reviewed):

- **5 low-severity:** every tracked sponsor shows real week-to-week score variance (coefficient of variation ranging from 0.80 for Busch Light to 1.42 for Cheddar's Scratch Kitchen). Expected, given that wins and playoff races create step-changes rather than smooth week-to-week visibility, and the smaller-sample sponsors (Cheddar's, Castrol) show the widest swings.

- **1 high-severity:** 3 race-sponsor combinations score at or near zero. These are races where a sponsor’s car had no laps led, no win, no playoff bonus, and no media mentions that week: the model’s theoretical floor, not a data quality issue.

2.4.2 Weight Sensitivity Testing

To test whether conclusions depend on the specific 40% race performance / 50% visibility index / 10% special events weighting, the model was rerun under four category-weight configurations:

Scenario	Race Performance	Visibility Index	Special Events
Baseline	40%	50%	10%
Performance Heavy	50%	40%	10%
Media Heavy	30%	60%	10%
Equal Weights	33.3%	33.3%	33.3%

Results: 4 of the 5 tracked sponsors held the exact same visibility rank in all four scenarios (Progressive #1, Castrol #4, Busch Light #5, Love’s Travel Stops last among the comparison groups). Cheddar’s Scratch Kitchen was the only tracked sponsor whose rank moved, swinging between #2 and #3 depending on how much weight a scenario gives to race performance versus media visibility. That’s consistent with its actual profile of strong Reddit/ YouTube engagement paired with minimal on-track dominance (39 laps led across 12 races, the second-lowest of the five tracked sponsors).

Caveat on reading this as “robustness”: the visibility gaps between sponsors are large (Progressive at 477.6 points vs. Love’s Travel Stops at 71.0), so almost any reasonable reweighting preserves the same broad ordering. Rank stability here demonstrates that the sponsor gaps are large, not that the model’s internal weight calibration is finely tuned. A smaller, more closely-matched sponsor set would likely show more rank movement under the same weight changes.

A second caveat specific to this test: the comparison set technically includes “Other/Unknown” (every untracked car aggregated into one bucket) alongside the five named sponsors. That bucket’s combined laps-lead total (12,895, versus 23-839 for any single tracked sponsor) pushes it into the #2-#3 rank range under performance-heavy weighting, despite representing no individual sponsor’s real visibility. Its presence in this specific ranking is a bookkeeping artifact of including untracked cars, not a sponsor comparison, and should be disregarded when reading the sensitivity results.

2.5 Assumptions and Limitations

Key Assumptions

Assumption	Implication	Mitigation
Social data represents fan engagement	Reddit/YouTube may not capture full audience	Acknowledged as proxy metrics
Cost estimates are accurate	Efficiency rankings depend on cost accuracy	Tested with cost ranges; see Section 2.4 sensitivity results
2025-2026 patterns are representative	Results may not generalize to other seasons	Framed as a two-season analysis, not a multi-year trend
News search captures relevant coverage	May miss some media mentions outside the tracked domain list	Used a consistent domain list and query methodology across all sponsors

Known Limitations

1. Cost Estimate Uncertainty Issue: Exact sponsorship costs are confidential. Our estimates are derived from industry publications and comparable-deal analysis, and may differ from actual contract values. This is most acute for Cheddar’s Scratch Kitchen, whose \$4.9M estimate borrows Castrol’s own sourced per-race rate rather than a disclosed figure for the Cheddar’s/RCR deal specifically.

Impact: Efficiency rankings could shift if actual costs differ significantly from estimates. Cheddar’s efficiency lead over Progressive is the primary recommendation’s single largest point of uncertainty.

Mitigation: Tested efficiency rankings with low, mid, and high cost estimates for every sponsor (see `sensitivity_analysis.md` and Section 2.4). Recommended next step: a direct quote or better-sourced comparable for the Cheddar’s deal.

2. Small Sample Sizes for Two of the Five Sponsors Issue: Castrol (11 races) and Cheddar’s Scratch Kitchen (12 races) have roughly a third of the tracked races that Love’s Travel Stops (29) has, and under half of Busch Light (25).

Impact: A single strong or weak week moves these two sponsors’ season totals substantially more than it would for the higher-race-count sponsors: both show the highest score volatility of the five ($CV = 0.90$ and 1.42 respectively, see Section 2.4 anomaly detection).

Mitigation: Flagged directionally rather than treated as precise point estimates; recommended revisiting the ranking once each has 10-15 more races on record.

3. Proxy Metrics for Visibility Issue: True “visibility” is difficult to measure directly. We use proxy metrics (mentions, views, finish position) rather than a direct visibility measurement.

Impact: The model captures correlates of visibility, not visibility itself.

Mitigation: Selected proxies with a demonstrated statistical relationship to performance (see Section 2.4 validation), and weighted the most reliable channel (Reddit) most heavily in that composite.

4. Two-Season Scope Issue: Analysis covers the 2025 season and 2026 season-to-date. Results may not apply to other seasons with different competitive dynamics, driver lineups, or sponsorship structures.

Impact: Cannot predict future sponsor performance from this data alone.

Mitigation: Framed conclusions as specific to the 2025-2026 window. Recommended validation with additional seasons before locking in a multi-year sponsorship decision.

5. Social Media Timing Issue: Social media metrics were collected at a point in time (July 9 - August 16, 2026) and may have changed since collection.

Impact: Engagement numbers represent a snapshot, not real-time values.

Mitigation: Collected data within a consistent window across all sponsors, so relative comparisons remain fair even if absolute counts have since moved.

6. Activation Not Measured Issue: Sponsorship value includes activation (hospitality, B2B networking, in-market promotions) that cannot be measured with public data.

Impact: Analysis captures visibility only, not total sponsorship ROI.

Mitigation: Clearly scoped this analysis as “visibility,” not full ROI.

What This Analysis Can and Cannot Tell You

This Analysis CAN Tell You	This Analysis CANNOT Tell You
Relative visibility rankings of the five analyzed sponsors	Whether sponsorship is a good investment overall
Which sponsors deliver the most visibility per estimated dollar	Actual sponsorship costs
How race performance correlates with visibility, and how strongly (Section 2.4)	Whether past visibility predicts future visibility
Which events (wins, playoffs, top-5s) generate the most visibility	How visibility translates to sponsor business outcomes

For questions about methodology, contact Lloyd Todaro (lloydltodaro@gmail.com).

Section 3: Findings

3.1 Finding: Efficiency Leadership Doesn't Track Brand Prestige

Cheddar's Scratch Kitchen (Kyle Busch, Richard Childress Racing) delivers the best visibility-per-dollar of any tracked sponsor, despite being a mid-tier team on a partial-season deal rather than a marquee, full-season primary.

Sponsor	Total Visibility	Visibility Rank	Efficiency (pts/\$M)	Efficiency Rank	Est. Cost
Progressive	477.6	#1	26.5	#2	\$18.0M
Cheddar's Scratch Kitchen	195.6	#2	39.9	#1	\$4.9M
Busch Light	177.0	#3	13.6	#5	\$13.0M
Castrol	108.3	#4	24.1	#3	\$4.5M
Love's Travel Stops	71.0	#5	15.8	#4	\$4.5M

Cheddar's is not a "cheap but small" pick. It's #2 in total visibility, trailing only Progressive, which costs roughly 3.7x as much. Its efficiency lead over the next-best sponsor (Progressive) is roughly 1.5x.

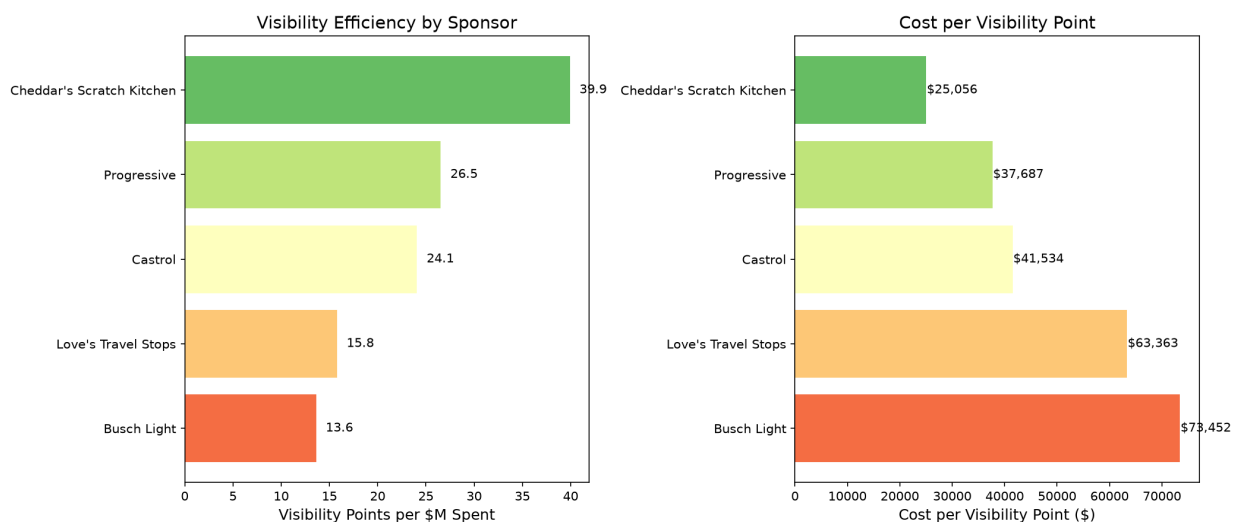


Figure 2: Visibility efficiency and cost per point by sponsor

The cost-per-point panel on the right makes the ranking concrete: Cheddar’s Scratch Kitchen costs about \$25,000 per visibility point, roughly a third of what Busch Light costs for the same unit of exposure (\$73,452) and well below Progressive’s \$37,687.

Caveat: the \$4.9M cost estimate behind this finding borrows Castrol’s own sourced per-race rate rather than a disclosed figure for the Cheddar’s/RCR deal specifically (see Section 2.4 and 2.5). This is the single largest open question behind the ranking.

3.2 Finding: A Playoff Race Is Worth More Than Double a Regular-Season One

The visibility bump from playoff races is not a marginal effect:

Event Type	Visibility Multiplier	Sample
Win vs. non-win	1.79x	All tracked races
Top-5 vs. outside-top-10 finish	3.12x	All tracked races
Playoff vs. regular-season race	2.29x (+129%)	20 playoff sponsor-entries vs. 77 regular-season entries

The playoff effect is a bigger swing than the win bonus and second only to the top-5 bonus, independent of finish position. Sponsors should treat playoff-window races (races 27-36) as prime, high-value activation windows, not just “more of the same, but with higher stakes.”

By contrast, “wins and top-5s generate more visibility than other finishes” is not, by itself, a useful headline finding: no one sponsoring a race car expects otherwise. Its value here is calibrating the scoring model’s special-events weighting (Section 2.3), not as a stakeholder-facing insight.

3.3 Finding: Reddit Is the Only Channel With a Statistically Reliable Signal

Of the three visibility channels tracked, only Reddit’s relationship with race performance clears standard statistical significance thresholds at this sample size:

Channel	Correlation with Finish Position	p-value	N	Significant?
Reddit mentions	$r = -0.34$	0.0006	97	Yes ($p < 0.001$)
YouTube views	$r = -0.20$	0.055	97	No (borderline)
News mentions	$r = -0.13$	0.20	97	No

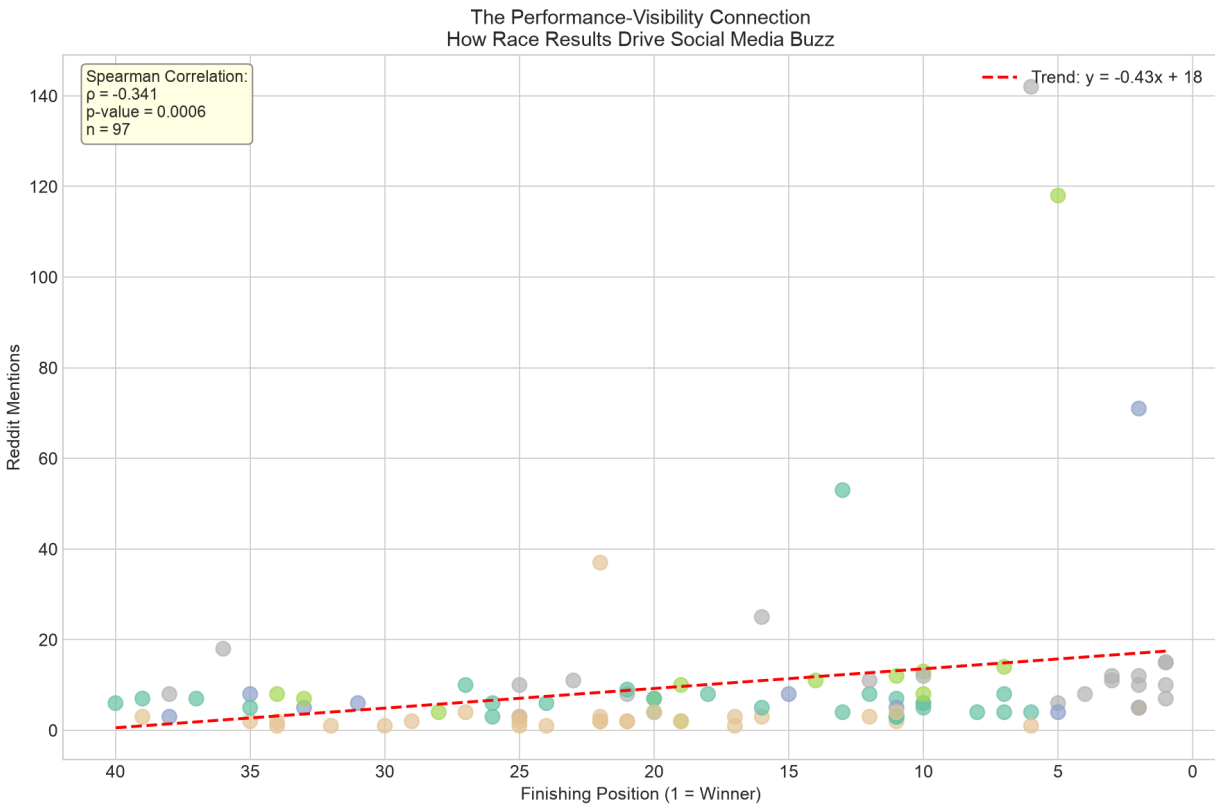


Figure 3: Race performance vs. Reddit mentions, with Spearman correlation and trend line

Most races cluster at single-digit mention counts regardless of finish position, the dense band along the bottom of the plot, and a small number of high-mention races pull the trend line into negative territory. That pattern, real but noisy rather than tight, is why Reddit clears significance at $p = 0.0006$ while YouTube ($p = 0.055$) and news ($p = 0.20$) don't: 97 races is enough to detect the underlying trend, but not enough to make it obvious point by point.

YouTube and news readings are directionally consistent with Reddit but not statistically distinguishable from noise at this sample size. This is a real constraint on how much weight either should carry in a future live monitoring dashboard. Reddit should be weighted most heavily, with YouTube and news treated as confirmatory rather than primary signals.

3.4 Finding: On-Track Performance and Visibility Move Together, But Not Proportionally

Sponsor	Avg. Finish	Wins	Top-5s	Top-10s	Total Visibility
Progressive	10.6	4	11	13	477.6
Cheddar's	17.5	0	1	4	195.6
Scratch Kitchen					
Castrol	18.9	0	3	3	108.3
Busch Light	18.7	0	-	-	177.0
Love's	21.7	0	-	-	71.0
Travel Stops					

Progressive is the only tracked sponsor with any wins in the window, and its average finish (10.6) is more than 7 spots better than the next-closest sponsor, consistent with its #1 visibility rank. But the relationship flattens out below that: Cheddar's, Castrol, and Busch Light post similar average finishes (17.5-18.9) yet materially different visibility totals (195.6, 108.3, 177.0), meaning on-track performance alone does not fully explain the visibility gap between them. This is the gap the PCA-weighted Visibility Index (Section 2.3) is designed to capture, and it's also why efficiency (Section 3.1), not raw visibility, is the more decision-relevant metric once cost is factored in.

Section 4: Recommendations

4.1 Primary Recommendation

Cheddar's Scratch Kitchen

Recommendation: Cheddar's Scratch Kitchen represents the strongest visibility-per-dollar opportunity for NY Racing, contingent on verifying its cost estimate before finalizing budget.

Why this recommendation:

1. **Best efficiency of the five tracked sponsors:** 39.9 visibility points per \$M invested, roughly 1.5x the next-best sponsor (Progressive).
2. **Not just a "cheap" pick:** #2 in total visibility (195.6 points), trailing only Progressive, despite the second-smallest race count (12).
3. **Lowest financial commitment of the top-2 visibility performers:** an estimated \$4.9M, versus Progressive's \$18.0M for the #1 spot.

Key metrics: - Visibility Score: 195.6 points (Rank #2 of 5) - Efficiency: 39.9 pts/\$M (Rank #1 of 5) - Estimated Investment: \$4.9M (range \$2.7M-\$7.1M, low-medium confidence)

Risk factors: - **Cost estimate is unverified:** the \$4.9M figure borrows Castrol's own sourced per-race rate rather than a disclosed number for this specific deal. If true cost runs \$7-8M, efficiency falls to roughly Castrol's range and the "clear leader" framing no longer holds. *Mitigation:* get a direct quote or better-sourced comparable before committing budget. - **Small sample, high volatility:** 12 races with the highest week-to-week score variance of the five sponsors ($CV = 1.42$). *Mitigation:* revisit the ranking once 10-15 more races are on record.

4.2 Alternative Recommendation

Progressive

Best for: Organizations prioritizing maximum total exposure and cost certainty over efficiency.

Why consider this alternative:

1. **Highest total visibility by a wide margin:** 477.6 points, driven by the strongest on-track performance of any tracked sponsor (4 wins, 11 top-5s in 20 races, a 55% top-5 rate).
2. **Best-sourced cost estimate of the group:** medium confidence, versus Cheddar's low-medium confidence.

Key metrics: - Visibility Score: 477.6 points (Rank #1 of 5) - Efficiency: 26.5 pts/\$M (Rank #2 of 5) - Estimated Investment: \$18.0M

Trade-off: roughly 3.7x Cheddar's estimated cost for a visibility-rank upgrade rather than an efficiency upgrade. Right choice if budget certainty and maximum reach matter more than return per dollar.

If Cheddar's cost verification comes back unfavorable, **Castrol** is the well-sourced middle option (see Section 4.3, Scenario C), a smaller sample (11 races) but a sourced, medium-confidence cost estimate and reasonable efficiency (24.1 pts/\$M).

4.3 Strategic Scenarios

Scenario A: Maximize Visibility

Metric	Value
Recommended Sponsor	Progressive
Visibility Score	477.6
Efficiency	26.5 pts/\$M
Investment Required	\$18.0M

Best for: Teams with budget flexibility prioritizing brand exposure

Scenario B: Maximize Efficiency

Metric	Value
Recommended Sponsor	Cheddar's Scratch Kitchen
Visibility Score	195.6
Efficiency	39.9 pts/\$M
Investment Required	\$4.9M

Best for: Teams prioritizing ROI justification

Scenario C: Balanced Approach / Hedge

Metric	Value
Recommended Sponsor	Castrol
Visibility Score	108.3
Efficiency	24.1 pts/\$M
Investment Required	\$4.5M

Best for: Teams wanting a defensible middle option with better-sourced cost data than Cheddar's, if Cheddar's/RCR isn't the right brand fit or its cost verification comes back unfavorable

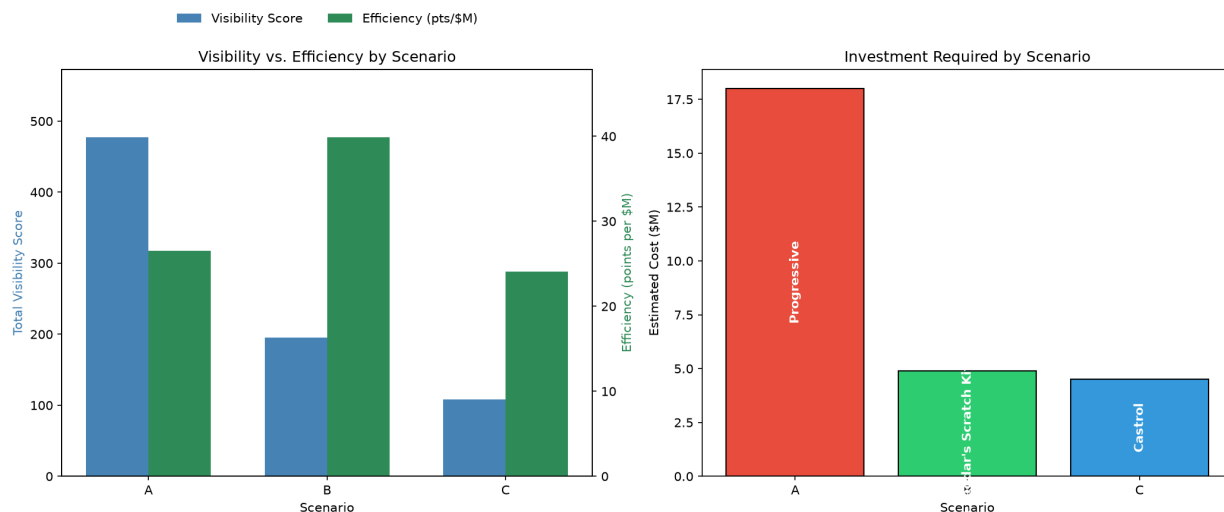


Figure 4: Scenario comparison: visibility and efficiency by strategic scenario

For NY Racing’s current situation, we recommend **Scenario B (Cheddar’s Scratch Kitchen)** as the lead option, with **Scenario C (Castrol)** as the fallback. See Section 4.1/4.2 for the full reasoning.

4.4 Implementation Guidance

Recommended Next Steps

1. **Immediate:** Request a direct cost quote (or a better-sourced comparable) for the Cheddar’s Scratch Kitchen / Kyle Busch / RCR No. 8 deal. It’s the single number that could flip this recommendation.
2. **Short-term (1-2 weeks):** Run a sentiment check on the Kyle Busch Reddit/YouTube corpus (`sentiment_analysis.ipynb`) to confirm the visibility is net-positive before pitching this externally.
3. **Medium-term (1 month):** Revisit the ranking once Cheddar’s and Castrol (both under 12 tracked races) have accumulated more races, to confirm the efficiency lead holds up beyond the current small sample.

Additional Research Recommended

Before finalizing any sponsorship decision, we recommend:

1. **Direct cost verification:** Obtain an actual cost quote or a genuinely comparable disclosed Cup deal for a similarly-profiled driver.
2. **Sponsor fixed-effects regression:** Test whether some sponsors carry a visibility premium independent of on-track performance, using data already in `scored_dataset.csv` / `master_dataset.csv` (no new data needed).
3. **Multi-season validation:** Re-run this ranking once additional seasons are available, to test whether the 2025-2026 pattern holds.
4. **Activation assessment:** Evaluate non-visibility benefits (hospitality, B2B networking) that this analysis does not capture (see Section 2.5).
5. **Brand fit analysis:** Assess alignment between sponsor brand and NY Racing’s values, particularly for Cheddar’s given Kyle Busch’s polarizing public profile.

4.5 Decision Framework

For future sponsorship evaluation, apply this framework:

Step	Action	Deliverable
1	Collect performance data	Race results dataset
2	Gather visibility metrics	Social + media data
3	Calculate visibility score	Normalized composite
4	Research cost estimates	Cost range table
5	Compute efficiency	Visibility per dollar
6	Run sensitivity test	Stability assessment
7	Make recommendation	Decision document

Future sponsorship decisions can reuse this same framework.

Appendix A: Full Data Tables

Final Rankings (Visibility and Efficiency)

primary_sponsor	total_visibility	visibility_rank	cost_mid	efficiency	efficiency_rank	rank_stability
Progressive	477.6	1	18	26.5	2	0
Cheddar’s Scratch Kitchen	195.6	2	4.9	39.9	1	1
Busch Light	177	3	13	13.6	5	0
Castrol	108.3	4	4.5	24.1	3	0
Love’s Travel Stops	71	5	4.5	15.8	4	0

Efficiency Rankings (Sorted, With Cost-per-Point and Confidence)

primary_sponsor	total_visibility	visibility_rank	cost_mid	efficiency	cost_per_point	efficiency_rank	confidence
Cheddar’s Scratch Kitchen	195.6	2	4.9	39.9	25056	1	low-medium
Progressive	477.6	1	18	26.5	37686.9	2	medium
Castrol	108.3	4	4.5	24.1	41533.8	3	medium
Love’s Travel Stops	71	5	4.5	15.8	63362.9	4	medium
Busch Light	177	3	13	13.6	73451.8	5	low-medium

Weight Sensitivity and Rank Stability

Sponsor	Min Rank	Max Rank	Rank Range
Progressive	1	1	0
Cheddar’s Scratch Kitchen	2	3	1
Castrol	4	4	0
Busch Light	5	5	0

Sponsor	Min Rank	Max Rank	Rank Range
Love's Travel Stops	6	6	0
Other/Unknown	2	3	1

Rank range = how much a sponsor's visibility rank moved across the four weight scenarios tested in Section 2.4.2 (baseline, performance-heavy, media-heavy, equal-weights). Only Cheddar's Scratch Kitchen (and the untracked "Other/Unknown" bucket) moved at all.

Appendix B: Methodology Details

Scoring Configuration

```
{
  "version": "1.0",
  "created_date": "2026-07-10",
  "category_weights": {
    "race_performance": 0.4,
    "visibility_score": 0.5,
    "special_events": 0.1
  },
  "variables": {
    "finish_position": {
      "category": "race_performance",
      "weight_within_category": 0,
      "effective_weight": 0,
      "direction": "inverse",
      "transform": "invert_position"
    },
    "laps_led": {
      "category": "race_performance",
      "weight_within_category": 1,
      "effective_weight": 0.4,
      "direction": "direct",
      "transform": "normalize"
    },
    "news_weighted_mentions": {
      "category": "visibility_score",
      "weight_within_category": "N/A",
      "effective_weight": "N/A",
      "direction": "direct",
      "transform": "normalize"
    },
    "reddit_mentions": {
      "category": "visibility_score",
      "weight_within_category": "N/A",
      "effective_weight": "N/A",
      "direction": "direct",
      "transform": "normalize"
    },
    "youtube_sponsor_views": {
      "category": "visibility_score",
      "weight_within_category": "N/A",

```

```

        "effective_weight": "N/A",
        "direction": "direct",
        "transform": "normalize"
    },
    "is_win": {
        "category": "special_events",
        "weight_within_category": 0.6,
        "effective_weight": 0.06,
        "direction": "direct",
        "transform": "binary"
    },
    "is_playoff": {
        "category": "special_events",
        "weight_within_category": 0.4,
        "effective_weight": 0.04,
        "direction": "direct",
        "transform": "binary"
    }
}
}
}

```

Weighting Rationale and Design Decisions

Why not equal weights?

Equal weights (14.3% each across the seven scored variables) would treat Reddit mentions as equally important as race wins, and would ignore the hierarchical importance of performance versus social engagement.

Why not data-driven weights from regression?

Regression-based weights would require a labeled “visibility outcome” variable, which doesn’t exist; would be sensitive to sample size and outliers, especially given the small samples for Castrol and Cheddar’s Scratch Kitchen; and would be harder to explain to stakeholders than a transparent, fixed weighting scheme. The hybrid approach used here (fixed category weights, with PCA determining the data-driven split *within* the visibility index, see Section 2.3) balances transparency against letting the data speak within the media-channel category.

Appendix C: Code Repository Guide

All analysis code is available in the project repository, in pipeline order:

File	Purpose
Scraping_For_Race_Results.ipynb	Scrapes race-by-race results and per-race primary sponsor data from Racing Reference
Clean_Race_Data.ipynb	Cleans scraped race data and assigns each race’s actual primary sponsor
Collect_Youtube_API_Data.ipynb	Collects YouTube video engagement data via YouTube Data API v3
reddit_arctic_shift_sentiment.ipynb	Collects Reddit mention data via the Arctic Shift API
News_Data_Collection_old.ipynb	Collects news article mentions via the GDELT DOC 2.0 API and BigQuery GKG
pytrends_data_collection.ipynb	Collects Google Trends interest data
data_merge_master.ipynb	Merges race, social, and news data into the master dataset

File	Purpose
<code>data_visualization.ipynb</code>	Builds the PCA-weighted Visibility Index and exploratory visualizations
<code>statistical_analysis.ipynb</code>	Runs correlation and significance testing (<code>correlation_results.csv</code>)
<code>sentiment_analysis.ipynb</code>	Sentiment analysis on the Reddit/YouTube text corpus
<code>ROI_Estimation.ipynb</code>	Computes visibility scores, cost estimates, efficiency rankings, and sensitivity testing
<code>Final_analysis.ipynb</code>	Compiles the report sections and this final report

End of Report